

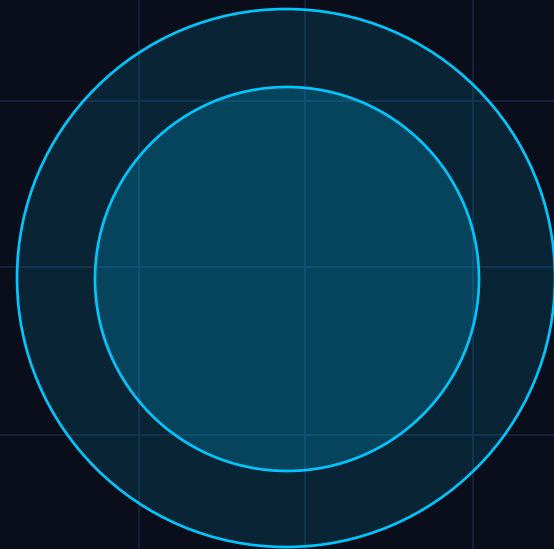
AUTOMATED SECURITY INTELLIGENCE

Pentrai

Automated Penetration Testing Platform

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4

Attack Phases

10+

Tools Integrated

Real-time

WebSocket Updates

AI

Risk Scoring

THE PROBLEM

Security Testing is Broken

\$4.88M

avg. data breach cost
(IBM 2024)



Too Slow

Manual pentesting takes weeks. Threats emerge in hours.



Too Expensive

Skilled pentesters cost \$200–\$400/hr, inaccessible to SMBs.



Inconsistent

Human error, scope gaps, and no standardized reporting.



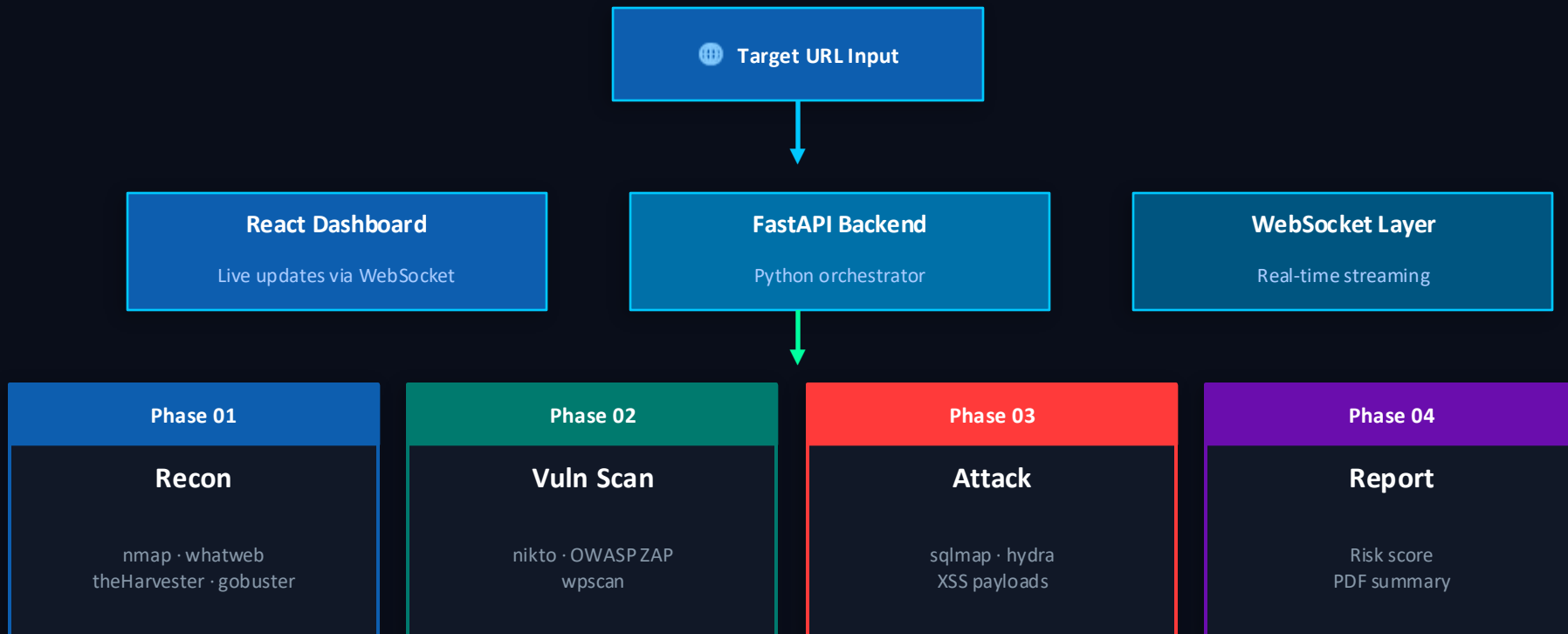
Not Continuous

Annual tests miss vulnerabilities introduced mid-cycle.

Organizations need automated, affordable, always-on security testing.

OUR SOLUTION

PentrAI — Full-Stack Attack Automation



WHO WE SERVE

Target Audience

SMBs & Startups

\$

Underserved segment

50–500 employees with no dedicated security team. Need affordable, automated scanning without hiring expensive consultants.

DevSecOps Teams

CI/CD

Integration-first

Engineering teams integrating security into CI/CD pipelines. Need continuous automated pentesting with programmatic APIs.

CTF & Security Researchers

500K+

Global community

Students, ethical hackers, and researchers who need a legal sandboxed environment to practice real-world attack scenarios.

Compliance Teams

Regulated

High-value contracts

Organizations needing PCI-DSS, SOC 2, ISO 27001 compliance evidence. Automated reports reduce audit prep by 70%.

How We Make Money

Starter

\$0
/mo

- ✓ 5 scans/month
- ✓ Recon + Vuln scan
- ✓ PDF report
- ✓ Community support

MOST POPULAR

Pro

\$9
/mo

- ✓ Unlimited scans
- ✓ All 4 attack phases
- ✓ API access + CI/CD
- ✓ Priority support

Enterprise

Custom

- ✓ On-premise deploy
- ✓ White-label reports
- ✓ SLA guarantee
- ✓ Dedicated CSM

System Architecture

React Frontend

- › URL Input Form
- › Live Scan Dashboard
- › WebSocket listener
- › Phase progress tracker
- › Findings explorer
- › Risk score display
- › PDF download

WebSocket



HTTP REST



FastAPI Backend

- › Scan orchestrator
- › Tool subprocess mgr
- › Phase state machine
- › Result aggregator
- › Risk scoring engine
- › WebSocket broadcaster
- › Report generator

Tool Pipeline

- › nmap
- › whatweb
- › theHarvester
- › gobuster
- › nikto
- › OWASP ZAP
- › wpscan
- › sqlmap
- › hydra
- › XSS engine

The 4-Phase Attack Pipeline



Build Timeline

Month 1–2

Foundation

 DONE

- › FastAPI skeleton + WebSocket
- › nmap & whatweb integration
- › Basic React dashboard
- › Docker environment setup

Month 3–4

Core Scanning

 DONE

- › Full Recon phase (4 tools)
- › Vuln scanning (nikto + ZAP)
- › wpscan CMS detection
- › Live result streaming

Month 5–6

Attack Modules

 IN PROGRESS

- › sqlmap integration
- › hydra brute-force module
- › XSS payload engine
- › Phase state machine

Month 7–8

Polish & Launch

 UPCOMING

- › PDF report generator
- › Risk scoring (CVSS)
- › User auth & API keys
- › Beta launch + feedback

Competitive Advantage

Feature	Pentrai	Manual Pentest	Burp Suite	Nessus
Fully Automated	✓	✗	⚠ Partial	⚠ Partial
Real-time Dashboard	✓	✗	✓	✗
AI Risk Scoring	✓	✗	✗	✗
WebSocket Streaming	✓	✗	✗	✗
Affordable (<\$100/mo)	✓	✗	⚠	✗
Open-source Tools	✓	⚠	✗	✗
Full Attack Chain	✓	✓	⚠	✗

Pentrai is the only solution combining full-chain automation + real-time UI + affordable pricing.